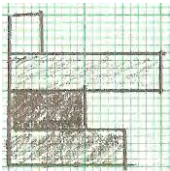


Question number	Answer	Additional guidance	Mark
4(a)	- surface area of one vein calculated (1) - total surface area = 1.4×10^3 / 1.5×10^3 (1) - length of vena cava = 40.7 / 40.8 / 42.7 (1)	37.68 DO NOT ACCEPT 37 ecf for wrong value in mp 1 but multiplied by 40 and incorrect standard form Bald answer in table or in space (provided same as in table if given) = 2 marks Bald answer of {36 / 37.68 / 37.7 / or in correct standard form} in box = 1 mark NB if they have shown the correct answer but not in standard form, award mp 1 as they must have calculated this value correctly (check the box below)	(3)

mp	Value used for π		
	3	3.14	calculator
1. S.A. of one vein	36	37.68	37.699111843077518861551720599354
2. Total S.A	1,440 1.4×10^3	1,507.2 1.5×10^3	1,507.9644737231007544620688239742 1.5×10^3
3. Length	42.666666666666666666666666666667 42.7	40.764331210191082802547770700637 40.8	40.743665431525205956834243423363 40.7

Question number	Answer	Additional guidance	Mark
4(b)	An explanation that includes the three of the following points: • blood in aorta is under higher pressure as it has just been pumped out of the {heart / (left) ventricle} (1) • aorta branches into many arteries / (blood flowing through) more vessels (1) • the {friction / resistance} between blood and vessel (slows the velocity down) (1) • blood cannot be pushed through because of {loss of elasticity in arteries / less elastic recoil} (1)	NB may need to piece together NB statements about the arteries do not need qualifying as the question is about arteries. If comparative statements about the aorta are made then they must be qualified ACCEPT velocity slower in arteries as they do not receive blood directly from the heart ACCEPT (total) cross-sectional area of arteries are greater than the aorta ACCEPT diameter of artery <u>lumen</u> is smaller / artery <u>lumen</u> is narrower diameter of aorta <u>lumen</u> is greater so velocity will be faster ACCEPT in the form of a comparison with the aorta being able to force blood because of more elastic fibres	(3)

ion number	Answer	Additional guidance	Mark
4(c)	An explanation that includes the following points: • more plasma is able to leave (the capillaries) (1) • results in faster diffusion (1) • more capillaries are in contact with the cells (in the tissue) (1)	ACCEPT named molecule e.g. oxygen, glucose ACCEPT faster gas exchange / rate of diffusion is fast enough / diffusion is rapid ACCEPT capillaries are close to more cells / diffusion distance is short	(2)

Question number	Answer	Additional guidance	Mark
4(d)(i)	The correct answer is A,  <i>B is incorrect because arteries have a high proportion of elastic fibres</i> <i>C is incorrect because arteries have an endothelial lining</i> <i>D is incorrect because artery walls contain collagen</i>		(1)

Question number	Answer	Additional guidance	Mark
4(d)(ii)	a diagram showing a bar for the endothelial cell layer only (1)	ACCEPT any length	(1)
